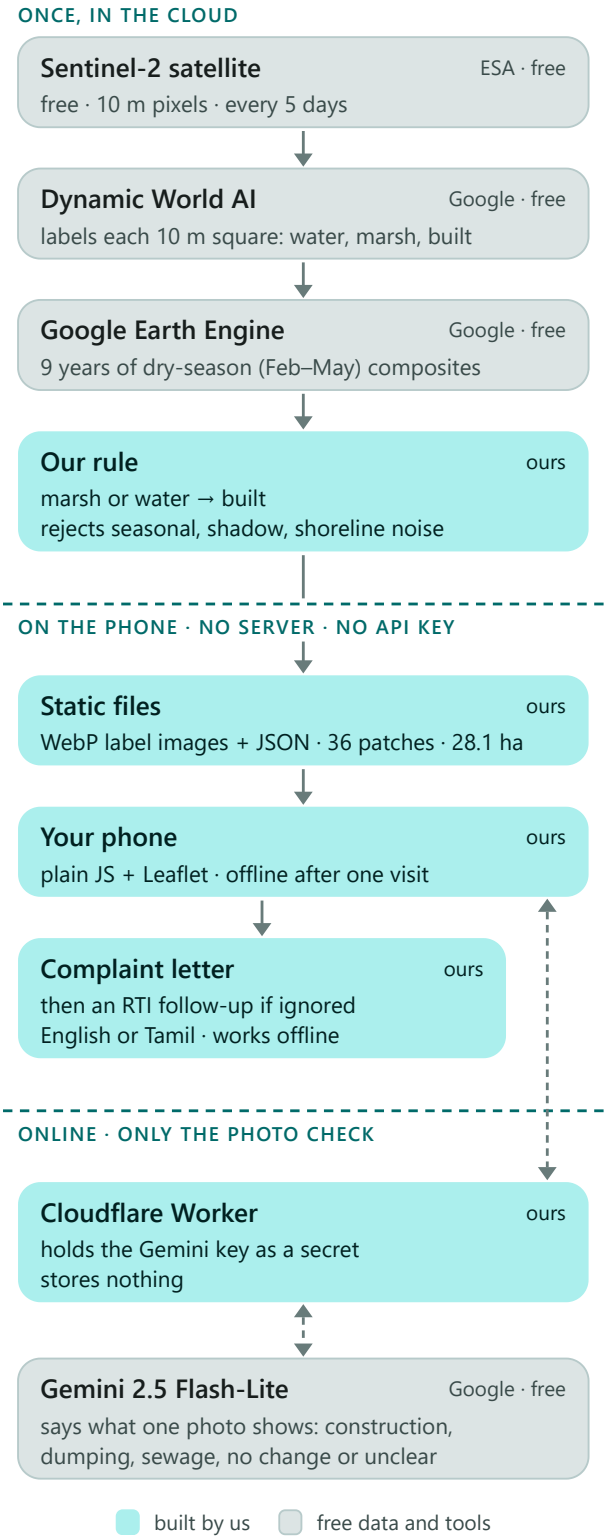
NEERKATTI

An AI water-keeper for Chennai's wetlands · Team Sternritters · DEXATHON 2026

28.1 ha	36	5	62
of marsh and water lost	patches flagged	live alerts	automated tests



WHAT IT DOES

Finds where Pallikaranai's marsh and water became buildings, season by season since 2018, from free Sentinel-2 images and Google's Dynamic World AI. Flags fresh filling as alerts, and writes a complaint letter in English or Tamil to the office that can act, then an RTI follow-up if it is ignored. A citizen's photo from the site can be checked by Gemini 2.5 Flash-Lite.

WHO THE LETTER GOES TO

- Tamil Nadu State Wetland Authority
- Greater Chennai Corporation
- Tamil Nadu Forest Department

WHY IT COSTS ₹0

The AI ran once, through Google Earth Engine's Python API. The website only reads the saved results: HTML, CSS and JavaScript, no server, no database. It works offline after one visit. The one live API call is the optional photo check, through our Worker, which holds the key.

LIMITS, SAID PLAINLY

- A 10 m pixel cannot see one house; small dumping is missed.
- Dynamic World is right about buildings 86.7% of the time but catches only 36.8% of marsh, so 28.1 ha is a floor, not a ceiling.
- Every patch is a possible change that needs a ground check. NEERKATTI never names an owner.
- The photo check needs a connection, and one photo read by an AI is not proof.
- The radar layer is not built yet.

Try it live
athulyapandian24.github.io/neerkatti/app/?final=1
Open on any phone. No sign-in.

62 automated tests run on every push · data: Copernicus Sentinel-2, Dynamic World (Brown et al. 2022), OpenStreetMap · code: github.com/ATHULYAPANDIAN24/neerkatti